Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Agentic AI · LangGraph · MCP · Multi-Agent Systems · RAG · LLM Evaluation · Cost Optimization · Python · AWS

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SUMMARY

AI engineer specializing in agentic AI architecture, multi-agent systems, LLM evaluation, and production AI operations. Built **Leasel** - an 8-agent LangGraph platform with tool/function calling, durable memory, MCP server integration, hybrid search RAG (ChromaDB, TF-IDF + semantic reranking), and HITL guardrails; CI eval harness with LLM-as-judge scoring (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLM evaluation and FinOps platform: cost-aware multi-provider routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; offline golden-dataset evals; 5-metric eval engine; CI regression gating. Currently at Intuit integrating Anthropic Claude into production financial services - RAG pipelines, structured outputs, guardrails, and evaluation harnesses. At Entrata (7+ years, 100K+ users), owned production systems end-to-end in a regulated multi-tenant SaaS platform. Strong across Python, LangGraph, MCP, REST APIs, Docker, Kubernetes, AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic AI Architecture · Multi-Agent Systems · LangGraph · LangChain · PydanticAI · MCP · Model Context Protocol · Tool Integrations · Tool-Use · Memory · Retrieval · Durable State · RAG Pipelines · Hybrid Search · Vector Databases · Embeddings · LLM Evaluation · LLM-as-Judge · Golden Datasets · CI Regression · AI Cost Optimization · FinOps · Multi-Provider Routing · Guardrails · Safety Controls · HITL · Responsible AI · Observability · Tracing · Structured Logging · OpenTelemetry · ML Pipelines · MLOps · scikit-learn · XGBoost · Python · FastAPI · TypeScript · Node.js · AWS (Lambda · Bedrock · SQS · API Gateway) · Docker · Kubernetes · Anthropic Claude · OpenAI · Azure OpenAI · Groq · Ollama · GitHub Actions CI/CD · Production Engineering · System Design

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **Production agentic AI:** Integrate Anthropic Claude into Python/FastAPI financial services - RAG pipeline with retrieval grounding, tool-use with durable state, structured-output schemas for downstream services, and prompt design for consistent sub-second responses.
- **Evaluation and quality:** Build LLM-as-judge evaluation harnesses with golden datasets and CI regression gates to catch hallucinations before release; implement guardrails with input validation, sensitive-data handling, and fallbacks; establish evaluation standards and document patterns for team reuse.
- Build ML data pipelines (scikit-learn, XGBoost) from ingestion through prediction APIs; containerize services with Docker through GitHub Actions CI/CD to AWS with CloudWatch monitoring.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned production backend systems end-to-end: HUD compliance ETL pipeline (180K records, zero violations across 6 years), async RabbitMQ event-driven processing with idempotency guarantees, PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement, 30% faster deployments); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual operational workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Agentic AI Platform: LangGraph · MCP · RAG · Evals

2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic AI platform: **8-agent LangGraph workflow** with tool/function calling, durable memory, and conditional routing; **MCP server integration** for shared tool access across agents; **hybrid search RAG** (OCR ingestion, section-aware chunking, TF-IDF + semantic reranking, citation grounding, ChromaDB); **HITL guardrails** routing regulated decisions to humans; **CI eval harness** with LLM-as-judge scoring: 93.6% precision, 87.9% F1 across 1,240 test cases. Three consuming surfaces share one retrieval and data layer.

AgentProbe - LLM Eval Framework · FinOps · Cost-Aware Multi-Provider Routing

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLM evaluation and FinOps platform: **offline evals** with golden datasets and LLM-as-judge; **online quality metrics** over time; **CI regression harness** gating releases on eval thresholds; **cost-aware multi-provider routing** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama - routes to cheapest model meeting quality bar; **5-metric eval engine** (accuracy, hallucination rate, retrieval quality, completeness, compliance); full tracing and structured logging across all agent runs.

AI on FHIR — Clinical NLP & LLM Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. HIPAA-style RBAC, OAuth-secured API integration, Docker-containerized. Sub-second response latency with retrieval grounding and structured output.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017